ALERT: A Landmark-Based, Fully Offline Driver Fatigue Monitor with Personalised Calibration and a Proposed Ensemble Classifier

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***Abstract*—Camera-based driver monitoring can warn of fatigue before a lapse becomes a crash, but cloud-hosted designs add network latency, fail without coverage, and send a driver's face off the vehicle. We present ALERT, a driver fatigue monitor that runs entirely on a consumer CPU. It reads facial landmarks from MediaPipe Face Landmarker and computes a nine-dimensional feature vector per frame: eye aspect ratio (EAR), its short-term deviation, a 60-second closure ratio (PERCLOS-style), blink rate, blink duration, mouth aspect ratio (MAR), yawn count, head pitch, and the ratio of MAR to EAR. A 10-second calibration at the start of each session sets driver-specific EAR and MAR thresholds, and a slow moving average tracks drift on long drives. Fatigue is graded on four levels, from alert to microsleep, rather than as a binary label. A working proof of concept (feature extraction, calibration, a Random Forest classifier, and a live overlay) is complete. Its classifier is trained only on synthetic data, so we report no accuracy figure as evidence of real-world performance. We also describe the planned production model, a two-layer bidirectional GRU with temporal attention for the UTA-RLDD dataset, and a proposed LightGBM–GRU–CNN ensemble. Neither has been trained on real footage. We state the known weaknesses of the current design, including thresholds that have not been tuned on real drivers, and list the experiments needed to resolve them.**

***Index Terms*—driver fatigue detection, driver monitoring systems, eye aspect ratio, PERCLOS, MediaPipe, on-device inference, temporal models**

I. Introduction

Drowsy driving is hard to measure and easy to underestimate. The U.S. National Highway Traffic Safety Administration (NHTSA) estimates that 91,000 police-reported crashes in 2017 involved a drowsy driver, with about 50,000 injuries and nearly 800 deaths, and cautions that the true figures are likely higher [1]. A 2024 AAA Foundation analysis of U.S. fatal crashes from 2017 to 2021 estimated that 17.6% involved a drowsy driver [2]. At 100 km/h a vehicle covers roughly 56 m in the two seconds a microsleep can last, with no one steering.

Regulators have begun to act. In the European Union, Regulation (EU) 2019/2144 requires a driver drowsiness and attention warning system in new vehicle types from July 2022 and in all newly registered vehicles from July 2024 [3]. Meeting such requirements with low-cost hardware is the setting ALERT is built for.

Mitigation approaches fall into three groups: vehicle-dynamics signals such as steering and lane position, physiological sensing such as EEG and ECG, and camera-based facial analysis. Vehicle-dynamics methods react to behaviour that has already degraded [4]. Physiological sensing is accurate but intrusive. A single RGB camera is passive and cheap, which is why we build on it.

ALERT contributes the following.

1) A fully offline pipeline from camera frames to a four-level fatigue grade, using a nine-feature temporal vector.
2) A 10-second per-session calibration of EAR and MAR thresholds with noise rejection and slow drift adaptation.
3) A documented path from a synthetic-data proof of concept to training on a naturalistic dataset, with the current limits of that proof of concept stated plainly.

Section II reviews related work, Section III states the gap, Section IV describes the method, Section V covers the implementation, Section VI reports proof-of-concept results and their limits, and Section VII discusses open problems.

II. Related Work

A. Eye closure measures

Soukupová and Čech defined the eye aspect ratio from six landmarks per eye and detected blinks from its temporal pattern [5]. PERCLOS, the proportion of time in a minute that the eyes are at least 80% closed, was established in driving-simulator work in 1994 [4] and later validated against psychomotor vigilance performance, after which FHWA and NHTSA described it as one of the most promising real-time alertness measures for in-vehicle systems [6], [7]. A 2023 review notes that definitions vary across studies (for instance in whether fast blinks are excluded, and in some work an EAR threshold stands in for eyelid opening), and that PERCLOS is less reliable for moderate drowsiness and in some real-driving conditions, which argues for combining it with other signals [8].

B. Landmark extraction

Kazemi and Sullivan's regression-tree method underlies the widely used 68-point dlib predictor [9]. MediaPipe provides a cross-platform framework for perception pipelines [10], and its face mesh model infers 468 3D landmarks from a single camera, with lighter variants for CPU inference [11]. The current Face Landmarker task returns 478 3D landmarks, the 468 mesh points plus iris points, along with optional blendshape scores and a facial transformation matrix [12].

C. Datasets and reported results

Public data are limited, and the choice of dataset shapes what a result means. The NTHU-DDD dataset contains infrared video of 36 participants simulating driving, in which drowsiness is acted [13], [14]. A deep-network system reached 73.06% on its evaluation set [15]. The UTA Real-Life Drowsiness Dataset (RLDD) contains about 30 hours of RGB video from 60 participants who were genuinely alert, low-vigilant or drowsy, recorded by the participants themselves on phones or webcams at under 30 frames per second [14]. Its authors note it was not recorded while driving and lacks night lighting and vehicle vibration. Their blink-sequence baseline reached 65.2% video-level accuracy, against 57.8% for human observers [14]. More recent work on RLDD combines features from the eyes, mouth, head pose and gaze with personalised thresholds and a soft-voting ensemble of Random Forest, XGBoost and an MLP, and reports 86.52% video accuracy [16]. That result is relevant to ALERT in two ways: it supports per-driver thresholds and multi-region features, and it shows that tree-based ensembles on hand-built features are competitive on this dataset.

D. Sequence models

Gated recurrent units [17] and additive attention [18] are standard tools for temporal classification. ALERT uses them as described in Section IV-E. LightGBM [19] is used for the tabular branch of the proposed ensemble.

III. Research Gap

1) **Coarse output.** Many systems emit a binary label. A mildly fatigued driver and one entering a microsleep call for different responses, so ALERT grades four levels.
2) **Fixed thresholds.** A single population EAR threshold is a poor fit when resting eye openness differs between drivers. Recent work on RLDD makes the same argument for per-driver thresholds [16]. ALERT calibrates per session.
3) **Single-channel fragility.** A landmark-only system degrades when landmarks are poor, for example under eyewear or low light. The proposed ensemble adds a temporal channel and a pixel-level channel. We have not yet measured whether this helps.
4) **Connectivity and privacy.** Offloading frames to a server adds latency, fails in dead zones, and moves face images off the vehicle. ALERT processes frames locally and persists only calibration thresholds.
5) **Data realism.** NTHU-DDD is acted and infrared, and RLDD is naturalistic but not recorded in a vehicle [14]. Neither fully represents in-cab conditions. We use RLDD as the primary training source and treat the gap to real driving as an open problem.

IV. Method

A. Pipeline overview

Figure 1 shows the five stages. Stages 1–3 and the display are implemented. Stage 4 runs a Random Forest trained on synthetic data in the proof of concept (POC). The production classifiers are specified in Section IV-E.

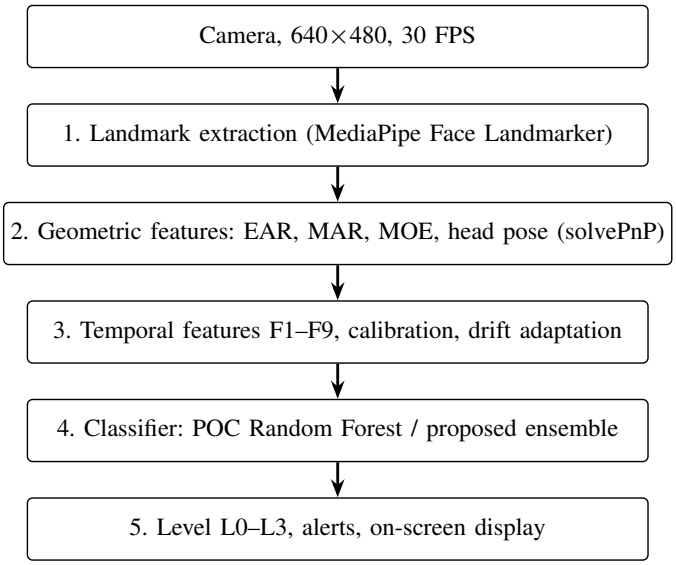


Fig. 1. ALERT processing pipeline.

B. Landmarks

Frames are captured at 640×480, 30 FPS, with a one-frame buffer to limit latency. MediaPipe Face Landmarker provides the 3D landmarks. ALERT uses fixed index subsets: six points per eye (left 33, 160, 158, 133, 153, 144; right 362, 385, 387, 263, 373, 380), eight inner-lip points (78, 81, 13, 311, 308, 402, 14, 82), and six anchors for head pose (nose tip 1, chin 152, eye corners 33 and 263, mouth corners 61 and 291).

C. Geometric features

With $p_1, \ldots, p_6$ the eye landmarks of [5],

$$\mathrm{EAR} = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\,\|p_1 - p_4\|}. \tag{1}$$

EAR is computed per eye and averaged. When the head is turned the eye is foreshortened and EAR falls even if the eye is open. For $|\psi| > 10^\circ$, where ψ is yaw from `solvePnP`, we apply a heuristic correction

$$\mathrm{EAR}_c = \frac{\mathrm{EAR}}{\cos\psi}, \qquad \cos\psi > 0.1. \tag{2}$$

This correction is geometrically motivated but has not been validated against ground truth. The mouth aspect ratio is defined over eight inner-lip points,

$$\mathrm{MAR} = \frac{\|p_2 - p_8\| + \|p_3 - p_7\| + \|p_4 - p_6\|}{3\,\|p_1 - p_5\|}, \tag{3}$$

and a combined signal is formed as $\mathrm{MOE} = \mathrm{MAR}/\mathrm{EAR}$, which rises when the mouth opens while the eyes narrow. Head pose uses a six-point canonical 3D face model with OpenCV `solvePnP`, intrinsics approximated from the frame size with focal length equal to frame width, and zero lens distortion. Pitch, yaw and roll are recovered by Rodrigues

TABLE I
FEATURE VECTOR COMPUTED PER FRAME

ID	Feature	Window
F1	EAR (corrected, smoothed)	per frame
F2	EAR standard deviation	5 s
F3	PERCLOS-style closure ratio	60 s
F4	Blink rate	60 s
F5	Mean blink duration (ms)	60 s
F6	MAR (smoothed)	per frame
F7	Yawn count	5 min
F8	Head pitch (deg)	per frame
F9	MOE = MAR/EAR	per frame

decomposition. Since the Face Landmarker can also output a facial transformation matrix [12], using it in place of a separate PnP step is a possible simplification.

D. Temporal features and calibration

Raw EAR and MAR are smoothed with a 5-frame moving average. Table I lists the nine features computed per frame.

PERCLOS-style closure ratio. Reference PERCLOS counts time at 80% eyelid closure or more [4], [7]. ALERT instead counts the fraction of frames in a 60-second window with EAR below the calibrated threshold. This is an approximation of the standard measure and should be reported as such.

Calibration. At session start the driver looks forward for 10 s (300 frames). Frames with EAR < 0.12 are discarded to exclude blinks. From the remaining samples,

$$T_{\mathrm{EAR}} = \mathrm{clamp}\big(\mu_E - 1.5\,\sigma_E,\ 0.15,\ 0.30\big), \tag{4}$$

$$T_{\mathrm{MAR}} = \mathrm{clamp}\big(\mu_M + 2.0\,\sigma_M,\ 0.40,\ 0.70\big). \tag{5}$$

If $\sigma_E > 0.06$ the calibration is treated as noisy and population defaults are used ($T_{\mathrm{EAR}} = 0.20$, $T_{\mathrm{MAR}} = 0.50$). Thresholds are stored locally in a small JSON file; no images are kept.

Drift adaptation. For long drives, a slow exponential moving average updates the EAR baseline,

$$\mu_E(t) = (1-\alpha)\,\mu_E(t-1) + \alpha\,\mathrm{EAR}(t), \quad \alpha = 0.001, \tag{6}$$

using only frames with $\mathrm{EAR} > 1.2\,T_{\mathrm{EAR}}$, so that closing eyes do not pull the baseline down. At 30 FPS the time constant is about 33 minutes.

Events. A blink is a dip of EAR below threshold that recovers, with duration accepted between 100 and 800 ms. A yawn is MAR above T_{MAR} for at least 2 s. For reference, a blink typically lasts 100 to 400 ms [5], so the upper bound of 800 ms deliberately admits slow closures as blinks, which should be revisited.

E. Fatigue levels

Table II gives the four levels. The PERCLOS bands are provisional engineering choices. They have not been derived from labelled data and are the first thing the real-data experiments should tune.

TABLE II
FATIGUE LEVELS AND PROVISIONAL BANDS

Level	Name	Closure ratio	Action
L0	Alert	< 0.08	monitor
L1	Mild	0.08–0.15	advisory chime
L2	Moderate	0.15–0.40	repeated alarm
L3	Severe	> 0.40, or eyes closed > 1.5 s	critical alarm

TABLE III
PLANNED TRAINING CONFIGURATION (NOT YET EXECUTED)

Item	Setting
Data	RLDD (primary), NTHU-DDD (robustness)
Split	cross-subject folds
Optimiser	AdamW, lr 10^{-3}, weight decay 10^{-4}
Schedule	cosine annealing, $T_{\max} = 50$ epochs
Loss, sampling	cross-entropy, inverse-frequency sampler
Stopping	patience 10 on validation loss; max 100 epochs
Batch size	64
Augmentation	Gaussian noise ($\sigma = 0.01$), frame dropout (10–20%), time scaling ($\pm 10\%$)

F. Classification

POC classifier. A Random Forest (30 trees, maximum depth 8) with standard scaling, trained on synthetic data, outputs four class probabilities. A rule overrides the output to L3 when head pitch is below -25° or the closure ratio exceeds 50%.

Production sequence model. For RLDD, whose labels are alert, low vigilant and drowsy [14], the production path uses a three-class head. It takes a 60-frame window of eight features and passes it through two stacked bidirectional GRU layers (64 and 32 units), a temporal attention layer that assigns one weight per time step, and two dense layers. The model has 63,939 parameters and a 0.25 MB footprint. Inference runs every 15 frames, about every 0.5 s at 30 FPS. The attention weights are drawn as a strip in the display, which shows which frames drove a prediction. Because RLDD videos are below 30 FPS, windows should be defined in seconds and resampled, not in frames.

Proposed ensemble. A second design combines three branches by weighted soft vote:

$$P = 0.50\,P_{\mathrm{LGBM}} + 0.30\,P_{\mathrm{GRU}} + 0.20\,P_{\mathrm{CNN}}. \tag{7}$$

The first branch is a LightGBM model [19] on the nine features. The second is a 32-unit GRU over a 5-second window of F1, F3, F5 and F8. The third is a small CNN on 16×32 grayscale eye crops. The weights are initial priors to be tuned on validation data. A hard rule forces L3 if the LightGBM L3 probability exceeds 0.85 or the eyes stay closed for more than 45 consecutive frames (1.5 s).

Planned training. Table III lists the configuration. It has not been run.

TABLE IV
SYNTHETIC CLASS PARAMETERS (MEAN, STD)

Class	EAR	Closure (%)	Blink dur. (ms)
L0	0.30, 0.025	5.0, 3.0	200, 30
L1	0.25, 0.020	20.0, 2.5	300, 40
L2	0.20, 0.018	32.0, 3.5	420, 50
L3	0.14, 0.020	55.0, 8.0	650, 120

V. Implementation

The POC lives in `POC_v0/`: a feature extractor (landmarks, EAR, MAR, MOE, pose), a preprocessor (smoothing, calibration, F1–F9, event tracking), a classifier wrapper with the L3 override, a script that generates synthetic data and trains the model, and an orchestrator that runs the live loop and display. The production code under `src/` contains the PyTorch model, a rolling-window real-time classifier, the calibrator, an alert state machine (normal, warning, critical) with audio, RLDD loading with cross-subject folds, feature caching, augmentation, a trainer and an evaluation script. The display shows state and confidence, live feature values, per-class probabilities, the attention strip, and a warm-up bar until the 60-frame buffer fills. Runtime memory is about 800 MB with MediaPipe, PyTorch and OpenCV loaded.

VI. Proof-of-Concept Evaluation

A. What this evaluation can and cannot show

The POC classifier is trained and tested on synthetic samples: 10,000 draws, 2,500 per class, from class-conditional Gaussians whose means were chosen from published PERCLOS-style ranges (Table IV). An 80/20 stratified split leaves 500 test samples per class. Because the classes are separated by construction, high accuracy here shows only that the pipeline and classifier run end to end. It says nothing about detecting fatigue in real drivers. We therefore do not present accuracy on this split as a result.

B. Held-out synthetic metrics

[Run `python POC_v0/train_model.py` and enter the per-class precision, recall, F1 and overall accuracy from its classification report here. Test support is 500 per class.]

C. Latency budget

The design budget per frame is about 12 ms for landmark extraction, 2 ms for geometric features, 1 ms for temporal features and under 2.5 ms for the Random Forest, which is well inside the 33 ms available at 30 FPS. These are planning figures. **[Replace with measured per-stage timings and the CPU model used.]**

D. Calibration check

Calibration was exercised on a synthetic landmark stream and completed at the expected frame. Under a Gaussian assumption, a threshold at $\mu - 1.5\sigma$ lies below about 93% of open-eye samples. The remaining 7% of open-eye frames would then count as closed, a baseline closure ratio near the 0.08 boundary between L0 and L1 in Table II. Real EAR distributions are not Gaussian and the blink filter removes some low samples, but this interaction needs to be measured and the multiplier or level bands adjusted.

VII. Limitations and Future Work

No real-data results. The most important gap is that nothing has been trained or evaluated on real footage. Training on RLDD with cross-subject folds is the next step. Results should be compared with the 65.2% baseline [14] and the 86.52% feature-ensemble result [16], and with attention to RLDD's self-reported labels and sub-30-FPS video.

Internal consistency of the synthetic data. In Table IV, the L1 closure mean of 20% sits in the L2 band of Table II, and L2 at 32% sits well inside it, so the synthetic generator and the provisional bands do not agree. One of them should be changed before any further tuning.

Ensemble. The LightGBM–GRU–CNN ensemble and its fusion weights are a design, not a measured improvement. An ablation against the single best branch is needed to justify the added cost.

Eyewear and low light. The current system needs visible-spectrum illumination, and landmark quality drops under occlusion. Candidate remedies are contrast-limited adaptive histogram equalisation [20], the pixel-level CNN branch, and a near-infrared camera path, which would require recalibration for the different spectral response.

Heuristic parts. The yaw correction of EAR, the -25° pitch override, the 800 ms blink bound and the closure bands are untested heuristics. Each should be evaluated individually.

Embedded deployment and alerts. The system has run only on a laptop CPU. Export through ONNX Runtime, OpenVINO or TensorRT is plausible but untested, and any latency or accuracy figures for those targets would need to be measured. Audio alerts work in software. Seat vibration or other haptic output needs hardware integration.

Safety. ALERT is a research prototype and must not be relied on as a safety-critical system.

VIII. Conclusion

ALERT is an offline fatigue monitor built from MediaPipe landmarks, a nine-feature temporal vector, per-session calibration and a four-level grade. The proof of concept works end to end, but it has only been trained on synthetic data and so supports no claim about real-world accuracy. The production sequence model and the proposed ensemble are specified and awaiting training. The next milestones are RLDD cross-subject training, tuning and validation of the level bands and calibration rule, measured latency, and tests under glasses and low light.

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